

programming language to get a taste of what an FPGA is all about.

Constituents of this design suite

There are three simulators that are used in the design suite: Deeds-DcS (digital-circuit simulator), Deeds-FsM (finite state machine) and Deeds-McE (microcomputer emulator). All these simulators can work simultaneously, allowing the design and simulation of complex networks, including a mix of standard logic, state machines and embedded microcomputers.

Deeds-DcS. The design of this suite provides a quasi-professional environment without compromising the comfort provided to the user. A graphical schematic editor is provided for the user interface (UI) along with a comprehensive library consisting of logical components. If the user has a finite machine component or a programmed microcontroller designed using the other two tools, he or she could include these in DcS library. Interaction with the digital circuit could be either through the editor directly or by using an event-driven timing simulator.

Deeds-FsM. Using FsM module, the user can easily describe and simulate the behaviour of an FsM. Also, evolution of inputs and outputs and the current state that are highlighted in the algorithmic state machine (ASM) chart could also be simulated. The schematic editor in Deeds-DcS can directly import files that are saved in Deeds-FsM as components.

Deeds-McE. A student can practice assembly-language-level programming with the help of this module. The emulated microcomputer system includes a custom 8-bit CPU, ROM and RAM memories, parallel input-output (I/O) ports, a reset circuitry and simple interrupt logic.

FPGA extension in Deeds

FPGA concepts are usually introduced to students during their final years of engineering. The reason behind this is that, they should have

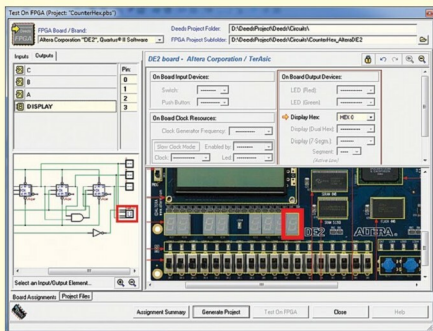


Fig. 2: Deeds-DcS allows users to export entire projects in VHDL language and test these on FPGA boards

